

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset8_expanded • Generated: 2026-04-14 20:23:38

I. CASE IDENTIFICATION	
Dataset ID	dataset8_expanded
Scan ID	ds-008-exp
Source File	windows_soc_data_8_expanded.csv
Primary Threat	Credential-Based Intrusion via Pass-the-Hash and LSA Secrets with Stealthy Persistence
MITRE Technique	T1059.001
Forensic Confidence	MEDIUM
Total Log Entries	10,000
Attacked Log Entries	155
Aggregate Risk Score	6,300
Threat Level	MEDIUM

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I EXECUTIVE SUMMARY

- A medium-severity, multi-stage intrusion campaign was identified across the environment, with a total of 155 confirmed malicious events out of 10,000 logs, resulting in an overall risk score of 63%.
- The attack primarily focused on credential theft and abuse, leveraging Pass-the-Hash (NTLM) and LSASS memory dumping techniques to gain and expand unauthorized access.
- Initial access and execution were achieved using PowerShell encoded commands and WMI-based remote execution, indicating the use of fileless and living-off-the-land techniques to evade detection.
- Attackers successfully authenticated using stolen NTLM hashes, bypassing the need for plaintext credentials and enabling stealthy lateral movement across systems.
- Credential harvesting was further intensified through LSASS memory dumps and extraction of LSA secrets, allowing attackers to obtain high-value credentials for privilege escalation.
- Persistence mechanisms included deployment of web shells on IIS servers and DLL side-loading via legitimate binaries, ensuring long-term access even after reboots or partial remediation.
- A malicious ASPX web shell was dropped and actively used for remote command execution, allowing attackers to run system-level commands such as 'whoami' and 'net user'.
- DLL side-loading attacks leveraged trusted applications like OneDrive updater to load malicious DLLs from non-standard paths, highlighting abuse of legitimate software for stealth persistence.
- Privilege escalation was achieved using token duplication techniques (CreateProcessWithToken), enabling attackers to spawn processes with SYSTEM-level privileges.
- Advanced defense evasion techniques were observed, including timestomping (modifying file timestamps) and environmental keying, where payloads execute only on specific target systems.
- Attackers performed reconnaissance by enumerating privileged domain groups such as Enterprise Admins and Schema Admins, identifying high-value targets within the network.
- Lateral movement was conducted using Windows Remote Management (WinRM), allowing attackers to execute commands on remote hosts and expand their control across multiple systems.
- Command and Control (C2) communication was established using external proxy techniques, including Cobalt Strike beacons over HTTPS with domain fronting to bypass network defenses.
- Data collection activities included continuous clipboard monitoring to harvest sensitive information, which was then written to disk and exfiltrated via HTTP POST requests to attacker-controlled infrastructure.
- A total of 8 distinct attack chains were identified, demonstrating a well-coordinated and structured intrusion campaign spanning multiple MITRE ATT&CK; tactics.
- The attack impacted 9 hosts and 11 user accounts, including privileged and service accounts, increasing the potential blast radius and risk of further compromise.
- The adversary demonstrated a high level of sophistication by chaining multiple techniques across execution, persistence, privilege escalation, defense evasion, and data exfiltration stages.
- Indicators suggest the use of advanced post-exploitation frameworks such as Cobalt Strike, commonly used in targeted attacks and red-team simulations.
- Despite the medium overall risk rating, the presence of critical techniques like Pass-the-Hash, LSASS dumping, and lateral movement significantly increases the likelihood of full domain compromise if not contained.
- Immediate response actions should include credential rotation, isolation of affected hosts, disabling NTLM where possible, and thorough investigation of privileged account activity.

- Long-term mitigation should focus on implementing least privilege access controls, enhancing endpoint detection and response (EDR), enabling PowerShell logging, and monitoring for anomalous authentication patterns.
- Overall, the attack represents a credential-centric intrusion campaign with strong emphasis on stealth, persistence, and lateral expansion, requiring urgent containment and strategic security improvements.

II

CASE IDENTIFICATION & BACKGROUND

Attack Narrative: Pass-the-Hash NTLM Authentication and LSASS Credential Dump ; Web Shell Persistence and LSA Secrets Dump ; DLL Side-Loading via Legitimate Updater Binary ; Token Duplication Privilege Escalation with Environmental Keying

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-DC-01, WIN-SOC-PROD-07, WIN-SOC-PROD-08
Users	NT AUTHORITY\SYSTEM, WIN-SOC\User1, WIN-SOC\Dev-01, WIN-SOC\WebSvc, WIN-SOC\SvcOps, WIN-SOC\ITAdmin, WIN-SOC\NetAdmin, WIN-SOC\DBAdmin

III

DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV

FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V

STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

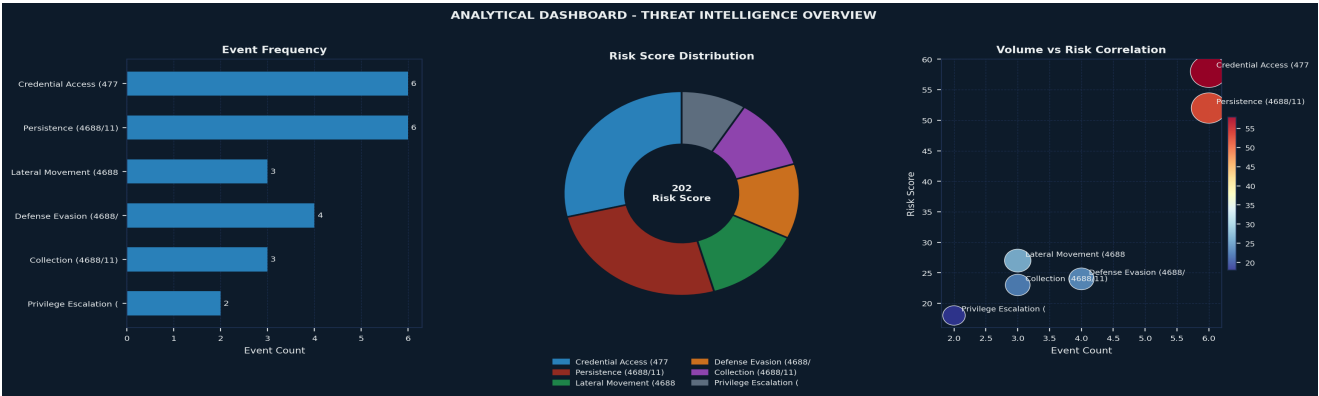


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

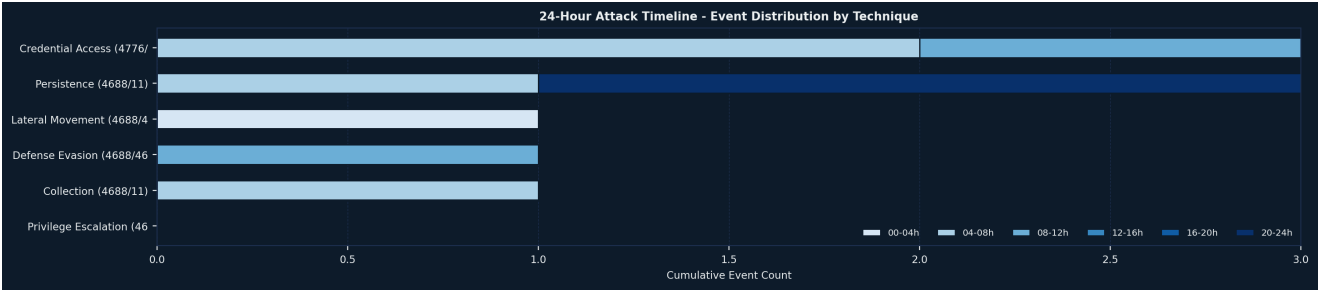


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (4776/4624)

LOW

Events: 6	Risk Index: 58	MITRE: T1550.002, T1003.001 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-04-03 00:19:33	WIN-SOC-PROD-02	WIN-SOC\Dev-01	4776
2026-04-03 00:19:34	WIN-SOC-PROD-02	WIN-SOC\User1	4624
2026-04-03 00:20:20	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-04-03 00:20:34	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10

Analysis: Credential Access (4776/4624) produced 6 suspicious artifacts with cumulative risk index 58. The mapped technique T1550.002, T1003.001 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688/4663)

LOW

Events: 4	Risk Index: 24	MITRE: T1480.001, T1070.006 Tactic: Defense Evasion
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Timestamp	Host	User Context	EID
2026-04-03 03:01:40	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688
2026-04-03 03:02:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688
2026-04-03 04:00:00	WIN-SOC-PROD-06	WIN-SOC\NetAdmin	4688
2026-04-03 04:00:45	WIN-SOC-PROD-06	WIN-SOC\NetAdmin	4663

Analysis: Defense Evasion (4688/4663) produced 4 suspicious artifacts with cumulative risk index 24. The mapped technique T1480.001, T1070.006 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4688/4624)

LOW

Events: 3		Risk Index: 27	MITRE: T1021.006 Tactic: Lateral Movement	
Timestamp		Host	User Context	EID
2026-04-03 05:00:00		WIN-SOC-PROD-07	WIN-SOC\ITAdmin	4688
2026-04-03 05:00:50		WIN-SOC-PROD-08	WIN-SOC\ITAdmin	4624
2026-04-03 05:01:40		WIN-SOC-PROD-08	WIN-SOC\ITAdmin	4688

Analysis: Lateral Movement (4688/4624) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1021.006 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688)

LOW

Events: 2		Risk Index: 18	MITRE: T1134.002 Tactic: Privilege Escalation	
Timestamp		Host	User Context	EID
2026-04-03 03:00:00		WIN-SOC-PROD-05	WIN-SOC\ITAdmin	4688
2026-04-03 03:00:50		WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Privilege Escalation (4688) produced 2 suspicious artifacts with cumulative risk index 18. The mapped technique T1134.002 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4688/11)

LOW

Events: 6	Risk Index: 52	MITRE: T1505.003, T1574.002 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-04-03 01:10:00	WIN-SOC-PROD-03	WIN-SOC\WebSvc	4688
2026-04-03 01:10:50	WIN-SOC-PROD-03	WIN-SOC\WebSvc	11

2026-04-03 01:11:40	WIN-SOC-PROD-03	WIN-SOC\WebSvc	4688
2026-04-03 02:00:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688

Analysis: Persistence (4688/11) produced 6 suspicious artifacts with cumulative risk index 52. The mapped technique T1505.003, T1574.002 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-08	7	6	MEDIUM
WIN-SOC-PROD-01	5	2	LOW
WIN-SOC-PROD-03	5	5	MEDIUM
WIN-SOC-PROD-05	4	3	LOW
WIN-SOC-PROD-06	4	0	LOW
WIN-SOC-PROD-04	3	2	LOW
WIN-SOC-PROD-02	2	2	LOW
WIN-SOC-DC-01	1	0	LOW
WIN-SOC-PROD-07	1	1	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Pass-the-Hash NTLM Authentication and LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

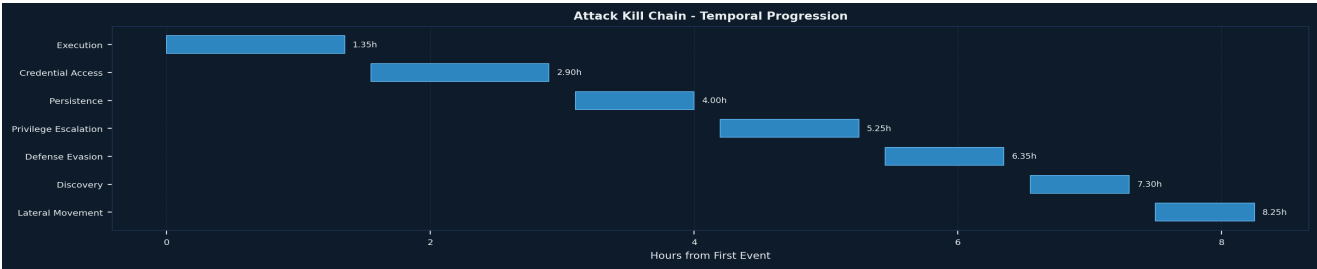
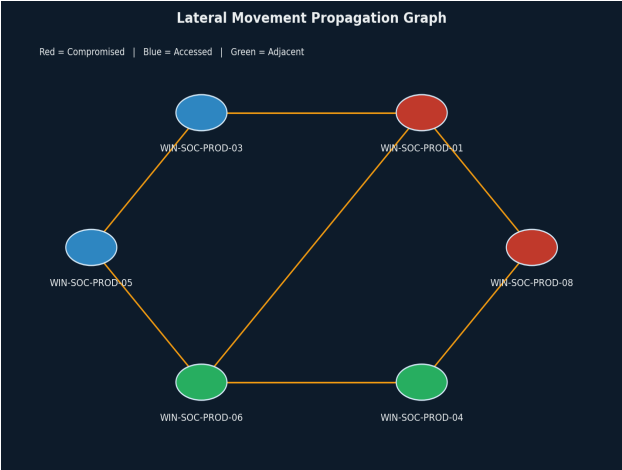


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-008-01	PowerShell Encoded IEX Download → WMI process call create → Pass-the-Hash NTLMv2 Authentication → Successful Logon Without Cleartext → LSASS Dump via mimikatz	T1059.001, T1047, T1550.002
chain-ds-008-02	IIS w3wp.exe Drops shell.aspx → Web Shell Remote Command Execution (whoami, net user) → LSA Secrets SECURITY Hive Saved → SYSTEM Hive Saved for Offline Decryption	T1505.003, T1505.003, T1003.004
chain-ds-008-03	OneDriveStandaloneUpdater.exe Loads Malicious version.dll → Unsigned DLL from Non-Standard Path → DLL Side-Load Payload Spawns Reverse Shell	T1574.002, T1574.002, T1574.002
chain-ds-008-04	CreateProcessWithToken Token Duplication → SYSTEM Context cmd.exe Confirmed → Environmental Keying Domain/Hostname Check → Payload Deployed Only on Matching Environment	T1134.002, T1134.002, T1480.001
chain-ds-008-05	Malware Timestamps Forged to Match System32 → MACE Attributes Overwritten → Enterprise Admins Group Enumerated → Schema Admins Group Enumerated → LDAP Privileged Group Query	T1070.006, T1070.006, T1069.002
chain-ds-008-06	Invoke-Command PowerShell Remoting to PROD-08 → WinRM Network Logon Accepted → wsmprovhost.exe Spawns Payload on Remote Host	T1021.006, T1021.006, T1021.006
chain-ds-008-07	Cobalt Strike Beacon via HTTPS to Redirector 103.56.52.10 → Domain Fronting via CDN to Evade Proxy Filtering	T1090.002, T1090.002
chain-ds-008-08	Get-Clipboard Polling Loop for Credential Harvesting → Harvested Data Written to clip_harvest.txt → Clipboard Data Exfiltrated via C2 HTTP POST to 103.56.52.10	T1115, T1115, T1115

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	wmic.exe	WMI Execution: wmic process call create powershell.exe -enc [payload]

Process	lsass.exe	Pass-the-Hash NTLM Authentication: Hash Reused from Stolen Credential
Event ID	4776	Security
Event ID	4624	Security
Process	mimikatz.exe	LSASS Dump Post Pass-the-Hash
Event ID	10	Sysmon
Process	w3wp.exe	IIS w3wp.exe Dropped Web Shell: C:\inetpub\wwwroot\shell.aspx
Event ID	11	Sysmon
Process	cmd.exe	Web Shell Remote Command Execution: Attacker Issued whoami and net user
Process	reg.exe	LSA Secrets Dump: reg save HKLM\SECURITY C:\Windows\Temp\sec.hiv

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

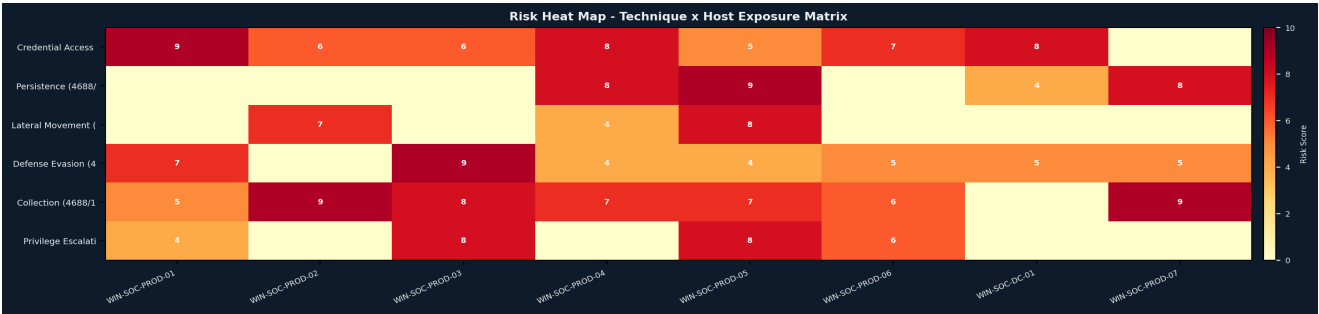


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

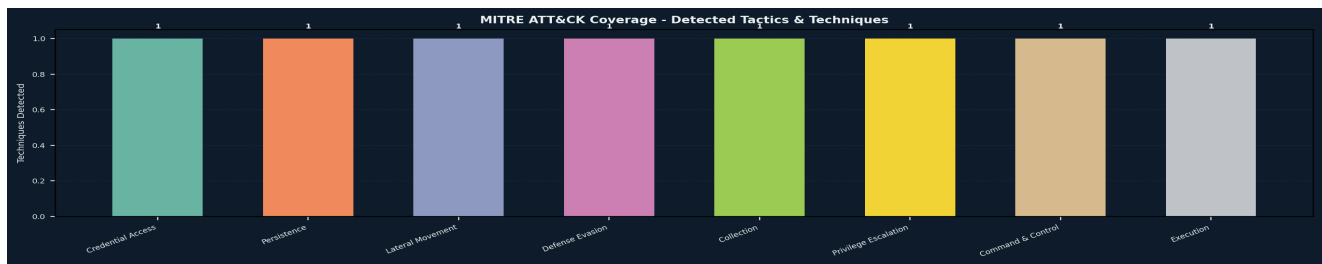


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1550.002, T1003.001	Credential Access (4776/4624)	Credential Access
T1505.003, T1574.002	Persistence (4688/11)	Persistence
T1021.006	Lateral Movement (4688/4624)	Lateral Movement
T1480.001, T1070.006	Defense Evasion (4688/4663)	Defense Evasion
T1115	Collection (4688/11)	Collection
T1134.002	Privilege Escalation (4688)	Privilege Escalation
T1090.002	Command & Control (5156/3)	Command & Control
T1059.001, T1047	Execution (4688)	Execution

XVII REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

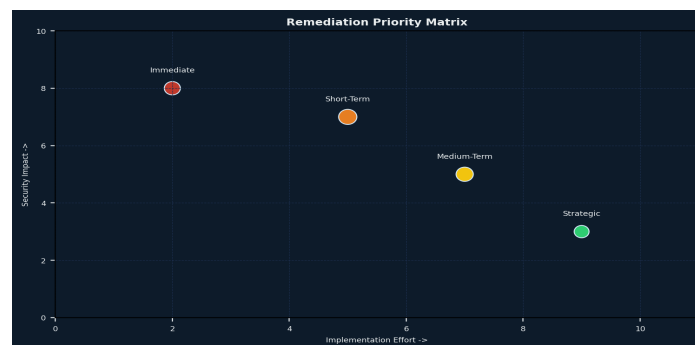


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset8_expanded**, the environment exhibits a **MEDIUM** severity posture with aggregate risk index **6,300** across **10,000** records. Dominant indicators include **Credential Access (4776/4624) (6 events)**, **Persistence (4688/11) (6 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst
Automated Forensic Engine

Authorizing Officer
Incident Response Division

XIX ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:23:40

XX ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-04-03 00:19:30	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-03 00:19:31	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-04-03 00:19:32	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	wmic.exe	WMI Execution: wmic process call create powershell.exe -enc [payload]
2026-04-03 00:19:33	WIN-SOC-PROD-02	WIN-SOC\Dev-01	lsass.exe	Pass-the-Hash NTLM Authentication: Hash Reused from Stolen Credential
2026-04-03 00:19:34	WIN-SOC-PROD-02	WIN-SOC\User1	lsass.exe	Pass-the-Hash Logon Success: NTLMv2 Type 3 from Non-Interactive Session
2026-04-03 00:20:20	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump Post Pass-the-Hash
2026-04-03 00:20:34	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-04-03 01:10:00	WIN-SOC-PROD-03	WIN-SOC\WebSvc	w3wp.exe	IIS w3wp.exe Dropped Web Shell: C:\inetpub\wwwroot\shell.aspx
2026-04-03 01:10:50	WIN-SOC-PROD-03	WIN-SOC\WebSvc	w3wp.exe	Web Shell File Written: shell.aspx contains cmd.exe execution handler

Timestamp	Computer	User	Process	Detail
2026-04-03 01:11:40	WIN-SOC-PROD-03	WIN-SOC\WebSvc	cmd.exe	Web Shell Remote Command Execution: Attacker Issued whoami and net user
2026-04-03 01:12:30	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	reg.exe	LSA Secrets Dump: reg save HKLM\SECURITY C:\Windows\Temp\sec.hiv
2026-04-03 01:13:20	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	reg.exe	LSA Secrets SYSTEM Hive Saved: reg save HKLM\SYSTEM C:\Windows\Temp\sys.hiv
2026-04-03 02:00:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	OneDriveStandaloneUpdater.exe	DLL Side-Loading: Malicious version.dll Loaded by Legitimate OneDrive Updater Binary
2026-04-03 02:00:50	WIN-SOC-PROD-04	WIN-SOC\SvcOps	OneDriveStandaloneUpdater.exe	Unsigned DLL Loaded: version.dll from Non-Standard Path C:\Users\SvcOps\AppData\Local
2026-04-03 02:01:40	WIN-SOC-PROD-04	WIN-SOC\SvcOps	powershell.exe	DLL Side-Load Payload Spawned Reverse Shell via PowerShell
2026-04-03 03:00:00	WIN-SOC-PROD-05	WIN-SOC\ITAdmin	powershell.exe	Token Duplication: CreateProcessWithToken Used to Spawn High-Privilege Shell
2026-04-03 03:00:50	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	cmd.exe	Elevated cmd.exe Spawned via Token Impersonation Confirming SYSTEM Context
2026-04-03 03:01:40	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	powershell.exe	Execution Guardrail Check: Domain Name and Hostname Validated Before Payload Deploy
2026-04-03 03:02:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	powershell.exe	Payload Executed Only After Confirming Target Environment Matches Attacker Criteria
2026-04-03 04:00:00	WIN-SOC-PROD-06	WIN-SOC\NetAdmin	powershell.exe	Timestomping: Malware File Timestamp Modified to Match System32 DLL Dates

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.